MATH 213

Formulas for standard error

Standard error for a sample proportion

The standard error for a sample proportion is

$$\sqrt{\frac{p(1-p)}{n}}$$

where *p* is the population proportion, and where *n* is the sample size.

Standard error for a sample mean

The standard error for a sample mean is

$$\frac{\sigma}{\sqrt{n}}$$

where σ is the population standard deviation, and where n is the sample size.

Problem!

If we knew p or σ we wouldn't need to compute confidence intervals at all since we would already know the population proportion or population standard deviation (and hence population mean).

Question: How do we use these formulas to ESTIMATE the standard error when we ONLY have sample data to work with?

We use the sample statistics as estimates for the population parameters. In other words, plug in $\hat{p}$ for p and s for σ. (There is a slight issue even with this for smaller sample sizes or very skewed sample data, but we'll talk about that)